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Migration of network traffic between licensed and unlicensed spectrum

Abstract

The present disclosure is directed to migrating network traffic from a licensed spectrum to an unlicensed spectrum within the same radio access technology (RAT). In one aspect, a method includes identifying a user device connected to a cellular wireless access technology, over a licensed spectrum; determining whether a condition for switching network traffic associated with the user device to an unlicensed spectrum is triggered; in response to determining that the condition is triggered, determining an unlicensed spectrum to move the network traffic to, the unlicensed spectrum being within a same cell as the licensed spectrum or in a different cell compared to a cell in which the licensed spectrum is; and migrating at least a portion of the network traffic to the unlicensed spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

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Background/Summary

RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/857,550, filed on Jul. 5, 2022, entitled, “MIGRATION OF NETWORK TRAFFIC BETWEEN LICENSED AND UNLICENSED SPECTRUM”, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The subject matter of this disclosure relates in general to the field of computer networking, and more particularly, to migrating user device traffic, applications traffic, user device traffic bound to certain network slices, or user device traffic in certain cell locations between a licensed spectrum and an unlicensed spectrum within the same radio access technology (RAT).

BACKGROUND

(2) Radio spectrum can be categorized into two types, a licensed spectrum and an unlicensed spectrum. A licensed spectrum is assigned exclusively to network operators for independent usage. As follows, licensed spectrum devices operate within the portion of the radio spectrum designated by the Federal Communications Commission (FCC) to be served for organizations that have been granted licenses. With exclusive rights, a license holder operates without interference in transmission. An unlicensed spectrum is assigned to every citizen for non-exclusive usage subject to some regulatory constraints. As follows, network operators can deploy cellular networks with more flexibility to manage interference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other advantages and features of the disclosure can be obtained, a more particular description of the principles briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only exemplary embodiments of the disclosure and are not, therefore, to be considered to be limiting of its scope, the principles herein are described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1A illustrates an example cloud computing architecture;

(3) FIG. 1B illustrates an example fog computing architecture

- (4) FIG. 2 depicts an exemplary schematic representation of a 5G network environment in which network slicing has been implemented, and in which one or more aspects of the present disclosure may operate;
- (5) FIG. 3 illustrates an example 5G network architecture with the deployment of two network slices according to some aspects of the present disclosure;
- (6) FIGS. 4A-4D illustrate an example flow of migrating network traffic from a network slice operating in a licensed spectrum to a network slice operating in an unlicensed spectrum, according to some aspects of the present disclosure;
- (7) FIG. 5 illustrates an example 5G network architecture with the deployment of a single network slice, according to some aspects of the present disclosure;
- (8) FIG. 6 illustrates an example flow of migrating network traffic from a licensed spectrum to an unlicensed spectrum within the same network slice, according to some aspects of the present disclosure;
- (9) FIG. 7 illustrates an example flow of reverting network traffic from an unlicensed spectrum to a licensed spectrum, according to some aspects of the present disclosure;
- (10) FIG. 8 illustrates a flow chart for an example method of migrating network traffic from a licensed spectrum to an unlicensed spectrum, according to some aspects of the present disclosure;
- (11) FIG. 9 shows an example computing system, which can be for example any computing device that can implement components of the system; AND
- (12) FIG. 10 illustrates an example network device, according to some aspects of the present disclosure.

DETAILED DESCRIPTION

- (13) Various embodiments of the disclosure are discussed in detail below. While specific implementations are discussed, it should be understood that this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without parting from the spirit and scope of the disclosure. Thus, the following description and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of the disclosure. However, in certain instances, well-known or conventional details are not described in order to avoid obscuring the description. References to one or an embodiment in the present disclosure can be references to the same embodiment or any embodiment; and, such references mean at least one of the embodiments.
- (14) Reference to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others.
- (15) The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Alternative language and synonyms may be used for any one or more of the terms discussed herein, and no special significance should be placed upon whether or not a term is elaborated or discussed herein. In some cases, synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and is not intended to further limit the scope and meaning of the disclosure or of any example term. Likewise, the disclosure is not limited to various embodiments given in this specification.
- (16) Without intent to limit the scope of the disclosure, examples of instruments, apparatus, methods and their related results according to the embodiments of the present disclosure are given below. Note that titles or subtitles may be used in the examples for convenience of a reader, which in no way should limit the scope of the disclosure. Unless otherwise defined, technical and

scientific terms used herein have the meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains. In the case of conflict, the present document, including definitions will control.

(17) Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be obvious from the description, or can be learned by practice of the herein disclosed principles. The features and advantages of the disclosure can be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. These and other features of the disclosure will become more fully apparent from the following description and appended claims or can be learned by the practice of the principles set forth herein.

Overview

(18) Disclosed herein are systems, methods, and computer-readable media for migrating network data traffic (network traffic) between a licensed spectrum and an unlicensed spectrum within the same radio access technology.

(19) In one aspect, a method includes identifying a user device connected to a cellular wireless access technology, over a licensed spectrum; determining whether a condition for switching network traffic associated with the user device to an unlicensed spectrum is triggered; in response to determining that the condition is triggered, determining an unlicensed spectrum to move the network traffic to, the unlicensed spectrum being within a same cell as the licensed spectrum or in a different cell compared to a cell in which the licensed spectrum is; and migrating at least a portion of the network traffic to the unlicensed spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

(20) In another aspect, the condition is a schedule of routing of network traffic of the user device between the licensed spectrum and the unlicensed spectrum, and the network traffic is migrated to the unlicensed spectrum according to the schedule.

(21) In another aspect, the condition is a capacity threshold of the licensed spectrum, and the network traffic is migrated to the unlicensed spectrum if the capacity threshold of the licensed spectrum is reached.

(22) In another aspect, the method further includes receiving, from a network element, a usage report that includes a volume of the network traffic of the user device to compare to the capacity threshold.

(23) In another aspect, the method further includes modifying a User Equipment Route Selection Policy (URSP) to migrate the portion of the network traffic to the unlicensed spectrum.

(24) In another aspect, the method further includes determining whether a second condition for switching the network traffic back to the licensed spectrum is met; and migrating the portion of the network traffic back to the licensed spectrum from the unlicensed spectrum.

(25) In another aspect, the method further includes transmitting a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the unlicensed spectrum.

(26) In one aspect, a network controller includes one or more memories having computer-readable instructions stored therein; and one or more processors. The one or more processors are configured to execute the computer-readable instructions to identify a user device connected to a cellular wireless access technology, over a licensed spectrum; determine whether a condition for switching network traffic associated with the user device to an unlicensed spectrum is triggered; in response to determining that the condition is triggered, determine an unlicensed spectrum to move the network traffic to, the unlicensed spectrum being within a same cell as the licensed spectrum or in a different cell compared to a cell in which the licensed spectrum is; and migrate at least a portion of the network traffic to the unlicensed spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

(27) In one aspect, one or more non-transitory computer-readable media include computer-readable instructions, which when executed by one or more processors of a network controller, cause the

network controller to identify a user device connected to a cellular wireless access technology, over a licensed spectrum; determine whether a condition for switching network traffic associated with the user device to an unlicensed spectrum is triggered; in response to determining that the condition is triggered, determine an unlicensed spectrum to move the network traffic to, the unlicensed spectrum being within a same cell as the licensed spectrum or in a different cell compared to a cell in which the licensed spectrum is; and migrate at least a portion of the network traffic to the unlicensed spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

Description of Example Embodiments

(28) The following acronyms are used throughout the present disclosure, provided below for convenience. AMF: Access and Mobility Management Function BNG: Broadband Network Gateway MBR: Modify Bearer Request NF: Network Function NRF: Network Repository Function PCF: Policy Control Function PDU: Protocol Data Unit RAN: Radio Access Network RAT: Radio Access Technology RRC: Radio Resource Control SMF: Session Management Function UDM: Unified Data Management UPF: User Plane Function USRP: UE Route Selection Policy

(29) Bandwidth refers to a measure of a bit rate of data communication resources, expressed in a number of bits communicated per unit time. Bandwidth throttling is a technique of reducing the speed at which data is communicated, which can be activated to limit network congestion in case of overcapacity either in a RAN or in a core network. Also, bandwidth throttling can be activated when subscriber usage exceeds a quota. Existing bandwidth throttling approaches utilize activating traffic-shaping rules on user plane functions such as UPF, and BNG. The net effect of bandwidth throttling results in creating two data pipelines, a fast lane and a slow lane. As a result, there can be some performance cost on the user plane function enforcing the artificial slow path rules. Further, these approaches can result in bad user experience for every subscriber using the application chosen for throttling. Therefore, there exists a need for an alternative approach to currently available bandwidth throttling techniques that can improve network traffic and congestion.

(30) As previously described, there are two types of radio spectrums: a licensed spectrum and an unlicensed spectrum. With the increased availability of unlicensed spectrum, network operators can deploy unlicensed spectrum as part of 3GPP access. As follows, a network operator can deploy both licensed spectrum and unlicensed spectrum. For example, a network operator can deploy a network slice in a cell operating in a licensed spectrum and the same network slice in another cell operating in an unlicensed spectrum. In another example, in a given cell, a network operator can operate a network slice in both licensed and unlicensed spectrums. Currently, offloading network traffic to unlicensed non-3GPP access at a RAT level is available. However, the existing approaches do not offer the mechanics of managing network traffic offloading between licensed and unlicensed spectrums within 3GPP access in view of enhancements to network slicing and the envisioned new slice configurations.

(31) Therefore, there exists a need for migrating network data traffic between a licensed spectrum and an unlicensed spectrum within the same RAT. The present technology includes systems, methods, and computer-readable media for solving the foregoing problems and discrepancies, among others. In some examples, systems, methods, and computer-readable media are provided for migrating users, applications, users bound to certain network slices, or users in certain cell locations between a licensed spectrum and an unlicensed spectrum within the same RAT. Further, the proposed solution relates to network traffic (e.g., back to the licensed spectrum) as the network traffic improves.

(32) In particular, the proposed solution can (1) migrate a session, a group of sessions, and certain applications from a licensed spectrum to an unlicensed spectrum (or vice versa) within the same network slice, (2) move a user between a cell supporting a given network slice in a licensed spectrum to another cell supporting the same network slice in an unlicensed spectrum, and (3) bind

such migration with various capacity threshold triggers.

(33) FIG. 1A illustrates a diagram of an example cloud computing architecture **100**. The architecture can include a cloud **102**. The cloud **102** can include one or more private clouds, public clouds, and/or hybrid clouds. Moreover, the cloud **102** can include cloud elements **104-114**. The cloud elements **104-114** can include, for example, servers **104**, virtual machines (VMs) **106**, one or more software platforms **108**, applications or services **110**, software containers **112**, and infrastructure nodes **114**. The infrastructure nodes **114** can include various types of nodes, such as compute nodes, storage nodes, network nodes, management systems, etc.

(34) The cloud **102** can provide various cloud computing services via the cloud elements **104-114**, such as software as a service (Saas) (e.g., collaboration services, email services, enterprise resource planning services, content services, communication services, etc.), infrastructure as a service (IaaS) (e.g., security services, networking services, systems management services, etc.), platform as a service (PaaS) (e.g., web services, streaming services, application development services, etc.), and other types of services such as desktop as a service (DaaS), information technology management as a service (ITaaS), managed software as a service (MSaaS), mobile backend as a service (MBaaS), etc.

(35) The client endpoints **116** can connect with the cloud **102** to obtain one or more specific services from the cloud **102**. The client endpoints **116** can communicate with elements **104-114** via one or more public networks (e.g., Internet), private networks, and/or hybrid networks (e.g., virtual private network). The client endpoints **116** can include any device with networking capabilities, such as a laptop computer, a tablet computer, a server, a desktop computer, a smartphone, a network device (e.g., an access point, a router, a switch, etc.), a smart television, a smart car, a sensor, a GPS device, a game system, a smart wearable object (e.g., smartwatch, etc.), a consumer object (e.g., Internet refrigerator, smart lighting system, etc.), a city or transportation system (e.g., traffic control, toll collection system, etc.), an internet of things (IoT) device, a camera, a network printer, a transportation system (e.g., train, motorcycle, boat, etc.), or any smart or connected object (e.g., smart home, smart building, smart retail, smart glasses, etc.), and so forth.

(36) The client endpoints **116** can communicate with the elements **104-114** as part of accessing network services through infrastructure intermediation messaging. Specifically, communications between the elements **104-114** and the client endpoints **116** can be managed and otherwise controlled through a network infrastructure between the client endpoints **116** and the cloud **102**. For example, any of a 5G infrastructure, an LTE infrastructure and a Wi-Fi infrastructure can communicate a physical location of a client endpoint to a cloud service. In turn, the cloud service can cause the infrastructure to send specific signaling to the client endpoint for accessing network services through the cloud service. For example, the cloud service can use the LTE infrastructure, e.g. through an LTE S1-M interface, to alert the client endpoint of Wi-Fi availability through the Wi-Fi infrastructure. In another example, the cloud service can use the Wi-Fi infrastructure, e.g. through MBO Wi-Fi messaging, to alert the client endpoint of LTE availability through the LTE infrastructure.

(37) FIG. 1B illustrates a diagram of an example fog computing architecture **150**. The fog computing architecture **150** can include the cloud layer **154**, which includes the cloud **102** and any other cloud system or environment, and the fog layer **156**, which includes fog nodes **162**. The client endpoints **116** can communicate with the cloud layer **154** and/or the fog layer **156**. The architecture **150** can include one or more communication links **152** between the cloud layer **154**, the fog layer **156**, and the client endpoints **116**. Communications can flow up to the cloud layer **154** and/or down to the client endpoints **116**.

(38) The fog layer **156** or “the fog” provides the computation, storage and networking capabilities of traditional cloud networks, but closer to the endpoints. The fog can thus extend the cloud **102** to be closer to the client endpoints **116**. The fog nodes **162** can be the physical implementation of fog networks. Moreover, the fog nodes **162** can provide local or regional services and/or connectivity

to the client endpoints **116**. As a result, traffic and/or data can be offloaded from the cloud **102** to the fog layer **156** (e.g., via fog nodes **162**). The fog layer **156** can thus provide faster services and/or connectivity to the client endpoints **116**, with lower latency, as well as other advantages such as security benefits from keeping the data inside the local or regional network(s).

(39) The fog nodes **162** can include any networked computing devices, such as servers, switches, routers, controllers, cameras, access points, gateways, etc. Moreover, the fog nodes **162** can be deployed anywhere with a network connection, such as a factory floor, a power pole, alongside a railway track, in a vehicle, on an oil rig, in an airport, in a shopping center, in a hospital, in a park, in a parking garage, in a library, etc.

(40) In some configurations, one or more fog nodes **162** can be deployed within fog instances **158**, **160**. The fog instances **158**, **160** can be local or regional clouds or networks. For example, the fog instances **158**, **160** can be a regional cloud or data center, a local area network, a network of fog nodes **162**, etc. In some configurations, one or more fog nodes **162** can be deployed within a network, or as standalone or individual nodes, for example. Moreover, one or more of the fog nodes **162** can be interconnected with each other via links **164** in various topologies, including star, ring, mesh or hierarchical arrangements, for example.

(41) In some cases, one or more fog nodes **162** can be mobile fog nodes. The mobile fog nodes can move to different geographic locations, logical locations or networks, and/or fog instances while maintaining connectivity with the cloud layer **154** and/or the endpoints **116**. For example, a particular fog node can be placed in a vehicle, such as a train, which can travel from one geographic location and/or logical location to a different geographic location and/or logical location. In this example, the particular fog node may connect to a particular physical and/or logical connection point with the cloud **154** while located at the starting location and switch to a different physical and/or logical connection point with the cloud **154** while located at the destination location. The particular fog node can thus move within particular clouds and/or fog instances and, therefore, serve endpoints from different locations at different times.

(42) FIG. 2 depicts an exemplary schematic representation of a 5G network environment **200** in which network slicing has been implemented, and in which one or more aspects of the present disclosure may operate. As illustrated, network environment **200** is divided into four domains, each of which will be explained in greater depth below; a User Equipment (UE) domain **210**, e.g. of one or more enterprise, in which a plurality of user cellphones or other connected devices **212** reside; a Radio Access Network (RAN) domain **220**, in which a plurality of radio cells, base stations, towers, or other radio infrastructure **222** resides; a Core Network **230**, in which a plurality of Network Functions (NFs) **232**, **234**, . . . , **n** reside; and a Data Network **240**, in which one or more data communication networks such as the Internet **242** reside. Additionally, the Data Network **240** can support SaaS providers configured to provide SaaSs to enterprises, e.g. to users in the UE domain **210**.

(43) Core Network **230** contains a plurality of Network Functions (NFs), shown here as NF **232**, NF **234** NF **n**. In some embodiments, core network **230** is a 5G core network (5GC) in accordance with one or more accepted 5GC architectures or designs. In some embodiments, core network **230** is an Evolved Packet Core (EPC) network, which combines aspects of the 5GC with existing 4G networks. Regardless of the particular design of core network **230**, the plurality of NFs typically executes in a control plane of core network **230**, providing a service based architecture in which a given NF allows any other authorized NFs to access its services. For example, a Session Management Function (SMF) controls session establishment, modification, release, etc., and in the course of doing so, provides other NFs with access to these constituent SMF services.

(44) In some embodiments, the plurality of NFs of core network **230** can include one or more Access and Mobility Management Functions (AMF; typically used when core network **230** is a 5GC network) and Mobility Management Entities (MME; typically used when core network **230** is an EPC network), collectively referred to herein as an AMF/MME for purposes of simplicity and

clarity. In some embodiments, an AMF/MME can be common to or otherwise shared by multiple slices of the plurality of network slices **252**, and in some embodiments an AMF/MME can be unique to a single one of the plurality of network slices **252**.

(45) The same is true of the remaining NFs of core network **230**, which can be shared amongst one or more network slices or provided as a unique instance specific to a single one of the plurality of network slices **252**. In addition to NFs comprising an AMF/MME as discussed above, the plurality of NFs of the core network **230** can additionally include one or more of the following: User Plane Functions (UPFs); Policy Control Functions (PCFs); Authentication Server Functions (AUSFs); Unified Data Management functions (UDMs); Application Functions (AFs); Network Exposure Functions (NEFs); NF Repository Functions (NRFs); and Network Slice Selection Functions (NSSFs). Various other NFs can be provided without departing from the scope of the present disclosure, as would be appreciated by one of ordinary skill in the art.

(46) Across these four domains of the 5G network environment **200**, an overall operator network domain **250** is defined. The operator network domain **250** is in some embodiments a Public Land Mobile Network (PLMN), and can be thought of as the carrier or business entity that provides cellular service to the end users in UE domain **210**. Within the operator network domain **250**, a plurality of network slices **252** are created, defined, or otherwise provisioned in order to deliver a desired set of defined features and functionalities, e.g. SaaSs, for a certain use case or corresponding to other requirements or specifications. Note that network slicing for the plurality of network slices **252** is implemented in end-to-end fashion, spanning multiple disparate technical and administrative domains, including management and orchestration planes (not shown). In other words, network slicing is performed from at least the enterprise or subscriber edge at UE domain **210**, through the RAN **220**, through the 5G access edge and the 5G core network **230**, and to the data network **240**. Moreover, note that this network slicing may span multiple different 5G providers.

(47) For example, as shown here, the plurality of network slices **252** include Slice **1**, which corresponds to smartphone subscribers of the 5G provider who also operates network domain, and Slice **2**, which corresponds to smartphone subscribers of a virtual 5G provider leasing capacity from the actual operator of network domain **250**. Also shown is Slice **3**, which can be provided for a fleet of connected vehicles, and Slice **4**, which can be provided for an IoT goods or container tracking system across a factory network or supply chain. Note that these network slices **252** are provided for purposes of illustration, and in accordance with the present disclosure, and the operator network domain **250** can implement any number of network slices as needed, and can implement these network slices for purposes, use cases, or subsets of users and user equipment in addition to those listed above. Specifically, the operator network domain **250** can implement any number of network slices for provisioning SaaSs from SaaS providers to one or more enterprises.

(48) 5G mobile and wireless networks will provide enhanced mobile broadband communications and are intended to deliver a wider range of services and applications as compared to all prior generation mobile and wireless networks. Compared to prior generations of mobile and wireless networks, the 5G architecture is service based, meaning that wherever suitable, architecture elements are defined as network functions that offer their services to other network functions via common framework interfaces. In order to support this wide range of services and network functions across an ever-growing base of user equipment (UE), 5G networks incorporate the network slicing concept utilized in previous generation architectures.

(49) Within the scope of the 5G mobile and wireless network architecture, a network slice comprises a set of defined features and functionalities that together form a complete Public Land Mobile Network (PLMN) for providing services to UEs. This network slicing permits for the controlled composition of a PLMN with the specific network functions and provided services that are required for a specific usage scenario. In other words, network slicing enables a 5G network operator to deploy multiple, independent PLMNs where each is customized by instantiating only

those features, capabilities and services required to satisfy a given subset of the UEs or a related business customer needs.

(50) In particular, network slicing is expected to play a critical role in 5G networks because of the multitude of use cases and new services 5G is capable of supporting. Network service provisioning through network slices is typically initiated when an enterprise requests network slices when registering with AMF/MME for a 5G network. At the time of registration, the enterprise will typically ask the AMF/MME for characteristics of network slices, such as slice bandwidth, slice latency, processing power, and slice resiliency associated with the network slices. These network slice characteristics can be used in ensuring that assigned network slices are capable of actually provisioning specific services, e.g. based on requirements of the services, to the enterprise.

(51) FIG. 3 illustrates an example 5G network architecture **300** with the deployment of two network slices according to some aspects of the present disclosure. Example 5G network architecture **300** comprises UE **302**, RAN **304**, a first slice **330A** operating in a licensed spectrum and a second slice **330B** operating in an unlicensed spectrum, and Internet **320**. Further, a cellular network with RAN **304** comprises a plurality of network functions such as AMF **306**, UE-PCF **308**, SM-PCF **310**, NS-ACF **312**, and UDM **314**. Each of the first slice **330A** operating in the licensed spectrum and the second slice **330B** operating in the unlicensed spectrum comprises multiple network functions such as SMF **332A**, UPF **334A**, PCF **336A**, NRF **338A** in the first slice **330A** and SMF **332B**, UPF **334B**, PCF **336B**, and NRF **338B** in the second slice **330B**.

(52) Referring to FIG. 2, the operator network domain **250** that provides cellular service to end-users in UE domain **210** (e.g., UE **302** as illustrated in FIG. 3) can deploy a plurality of slices, Slice **1**, Slice **2**, Slice **3**, and Slice **4** (e.g., first slice **330A** and second slice **330B** as illustrated in FIG. 2) where the network slicing can be performed from at least the enterprise or subscriber edge at UE domain **210**, through RAN **220** (e.g., RAN **304** as illustrated in FIG. 3), through the 5G access edge and 5G core network **230**, and to the network **240** (e.g., Internet **320** as illustrated in FIG. 3).

(53) In some examples, RAN **304** can support both the first cell operating in the licensed spectrum (e.g., 1800 MHz) and the second cell operating in the unlicensed spectrum (e.g., 3.5 GHz). For example, a network operator can deploy two network slices, first slice **330A** (e.g., Internet-L) and second slice **330B** (e.g., Internet-UL), for the same service. More specifically, a network operator can have two cells, one is a licensed cell (e.g., cell-L) and another an unlicensed cell (e.g., cell-UL). As follows, first slice **330A** (e.g., Internet-L) can be attached to the licensed cell (e.g., cell-L) and second slice **330B** (e.g., Internet-UL) can be attached to the unlicensed cell (e.g., cell-UL). Any time the network operator throttles its service, the network operator can change the URSP for a user device and all (or a portion of) traffic can be directed to the unlicensed slice (i.e., unlicensed cell). As this approach results in changing the network slice, there can be PDU impact (i.e., session continuity).

(54) In some examples, network data traffic that is transmitted to the licensed spectrum (i.e., first cell **330A**) can be moved/migrated to the unlicensed spectrum (i.e., second cell **330B**), which can be activated by various triggers. According to some examples, migration of the network traffic can be initiated by various triggers. Examples of triggers can include, but are not limited to, the following conditions: (1) a subscriber has exceeded its quota or violated fair usage policy (FUP); (2) RAN is experiencing a heavy load on a licensed spectrum; (3) the network slice in operation has hit the maximum per-slice-MBR threshold; (4) a given subscriber has hit a per-UE-per-slice-MBR threshold; (5) a given subscriber has hit a per-slice PDU Session Count limit; and (6) User Plane hits maximum forwarding capacity limits.

(55) FIGS. 4A-4D illustrate an example flow **400** of migrating network traffic from a network slice operating in a licensed spectrum to a network slice operating in an unlicensed spectrum according to some aspects of the present disclosure. Similar to example 5G network architecture **300** as illustrated in FIG. 3, network environment for example flow **400** in FIGS. 4A-4D comprises UE **302**, gNodeB (i.e., RAN) **304**, AMF **306**, SMF **332A** of the licensed spectrum, UPF **334A** of the

licensed spectrum, SMF 332B of the unlicensed spectrum, UPF 334B of the unlicensed spectrum, UE-PCF 308, SM-PCF 310, NS-ACF 312, and UDM 314.

(56) FIG. 4A, in particular, illustrates the process of UE registration and the first example of a trigger (Trigger 1) for the migration of network traffic. According to some examples, at step 402, a network operator can have gNodeB 304 that supports both a first cell operating in a licensed spectrum on F1 radio frequency and a second cell operating in an unlicensed spectrum on F2 radiofrequency.

(57) At step 404, UE 302 starts an application, which then according to a URSP associated with the application, can select a network cell operating in a licensed spectrum. As follows, at step 406, UE 302 connects to the network cell operating in the licensed spectrum on F1 radiofrequency.

(58) At step 408, UE registration is completed between UE 302 and AMF 306. At step 410, AMF 306 and UDM 314 confirm the subscription of UE 302 and process authentication of UE 302. At step 412, UE 302 sends a request to establish a PDU Session to AMF 306 via gNodeB 304, which then sends a PDU Session response to UE 402.

(59) In some examples, the migration of network traffic from a licensed spectrum to an unlicensed spectrum can be triggered when a per-slice PDU session exceeds a slice capacity. More specifically, at step 414, UE 302 sends an Admission Control request to NS-ACF 312. At step 416, NS-ACF 312 determines that the current slice PUD session has exceeded a predefined slice capacity. As follows, at step 418, NS-ACF 312 transmits, to AMF 306, an Admission Control Response with a rejection, which would trigger AMF 306 for the migration of the network traffic. For example, at step 420, AMF 306 is triggered to move UE 302 to the unlicensed spectrum. In some examples, a portion of the network traffic of UE 302, if not all, can be migrated to the unlicensed spectrum.

(60) FIG. 4B is a continuous diagram of FIG. 4A. In particular, FIG. 4B illustrates the second example of the trigger (Trigger 2) for the migration of network traffic. At step 422, AMF 306 sends a request to establish a PDU Session to SMF 332A/UPF 334A, which then sends an SM policy to SM-PCF 310 at step 424. At step 426, SM-PCF 210 determines that a per-slice-MBR quota would be exceeded if a new PDU session is admitted. As follows, at step 428, SM-PCF 310 sends a rejection message to SMF 332A/UPF 334A. Thus, at step 430, SMF 332A is triggered to move UE to the unlicensed spectrum. In some examples, a portion of the network traffic of UE 302, if not all, can be migrated to the unlicensed spectrum.

(61) FIG. 4C is a continuous diagram of FIG. 4B. In particular, FIG. 4C illustrates the process of the migration of network traffic activated by two different examples of triggers (e.g., Trigger 3 and Trigger 4). At step 432, the PDU establishment is accepted between UE 302 and SMF 332A/UPF 334A. As a result, at step 434, network traffic associated with UE 302 is directed to SMF 432A/UPF 434A.

(62) At step 436, AMF 306 can be triggered to migrate, to the unlicensed spectrum, a portion of the network traffic associated with UE 302 based on a predefined schedule (e.g., at 4 pm or any other configurable time-based trigger or schedule). Further, the schedule can define which session/application or an amount of the network data to be migrated to the unlicensed spectrum.

(63) In some examples, at step 438, SMF 332A/UPF 334A of the licensed spectrum can send a usage report to SM-PCF 310. The usage report can include a volume of network data that has been used by UE 302. At step 440, SM-PCF 310 can trigger SMF 332A to move UE 302 to the unlicensed spectrum if the UE volume of network data has exceeded a quota threshold.

(64) As shown in FIGS. 4A-4C, any of Trigger 1, Trigger 2, Trigger 3, or Trigger 4 can cause the Session of UE 302 to move from the licensed spectrum to the unlicensed spectrum at step 442. While four example triggers are described with reference to FIGS. 4A-4C, the present disclosure is not limited to these four examples and any other trigger for moving network traffic from a licensed spectrum to an unlicensed spectrum may fall within the scope of the present disclosure.

(65) FIG. 4D is a continuous diagram of FIG. 4C. At step 444, SM-PCF 310 can decide to move

UE **302** to the unlicensed spectrum. At step **446**, SM-PCF **310** sends an SM policy update message, to SMF **332B**/UPF **334B** of the unlicensed spectrum, regarding the migration of the session to the unlicensed frequency. At step **448**, SMF **332B**/UPF **334B** of the unlicensed spectrum sends an N1/N2 message to AMF **406** regarding the same.

(66) In some examples, at step **452**, SM-PCF **310** sends a UE Policy Update message to AMF **306**, which then sends a UE configuration update to UE **302** (e.g., updated URSP) at step **452**. At step **454**, UE **302** can start the application, which then according to the updated URSP, selects a network cell operating in a licensed spectrum. At step **456**, UE **302** sends a PDU Session Release message (e.g., to reconnect) to SMF **332B**/UPF **334B** of the unlicensed spectrum. As follows, at step **458**, UE **302** connects to the network cell operating in the unlicensed spectrum on F2 radiofrequency. At step **460**, a PDU Session Release is established between UE **302** and UE-PCF **308**. Further, at step **462**, the network traffic of UE **302** is transmitted on the unlicensed spectrum.

(67) FIG. 5 illustrates an example 5G network architecture **500** with the deployment of a single network slice, according to some aspects of the present disclosure. Example 5G network architecture **500** comprises UE **502** (can be the same as UE **302**), gNodeB **504** (can be the same as gNodeB **304**), network slice **506**, and Internet **520**. Further, network slice **506** can have a plurality of associated network functions such as AMF **508**, SMF **510**, UPF **512**, PCF **514**, UDM **516**, NRF **518**, etc.

(68) In some examples, network slice **506** can operate in both a licensed spectrum (e.g., 1800 MHz) and an unlicensed spectrum (e.g., 2.5 GHz). More specifically, a network operator can have one slice, which operates on both licensed and unlicensed frequencies. As follows, any time the network operator throttles its service, a user device can switch from a licensed spectrum to an unlicensed spectrum based on an RRC reconfiguration. Under this approach, since the network slice does not change, there is no PDU impact. Further, in some examples, UE **502** can have an application that must be on a licensed spectrum and another application that must be on an unlicensed spectrum.

(69) FIG. 6 illustrates an example flow **600** of migrating network traffic from a licensed spectrum to an unlicensed spectrum within the same network slice, according to some aspects of the present disclosure. Similar to example 5G network architecture **500** as illustrated in FIG. 5, network environment for example flow **600** in FIG. 5 comprises UE **502**, gNodeB **504**, AMF **508**, SMF **510**, UPF **512**, SM-PCF **514**, and UDM **516**.

(70) According to some examples, gNodeB **504** can be configured to support both licensed and unlicensed frequencies (e.g., cell-L on F1 radiofrequency and cell-UL on F2 radiofrequency) at step **605**. At step **610**, UE **502** connects to cell-L on F1 radiofrequency via gNodeB **504**.

(71) At step **615**, UE registration is completed between UE **502** and AMF **508**. At step **620**, AMF **508** and UMD **516** confirm the subscription of UE **502** and process authentication of UE **502**. At step **625**, UE **502** sends a PDU request (i.e., a request to establish a PDU session) to SMF **510**/UPF **512**, which then sends an SM policy message to SM-PCF **514** at step **630**.

(72) At step **635**, network traffic associated with UE **502** is directed to SMF **510**/UPF **512**. At step **640**, SMF **510**/UPF **512** prepares and sends a usage report to SM-PCF **514**. In some instances, the usage report can include a volume of network data associated with UE **502**. At step **645**, SMF **510**/UPF **512** determines that the volume of network data associated with UE **502** has exceeded a quota threshold. As follows, at step **650**, SM-PCF **514** decides to move UE **502** to an unlicensed spectrum (i.e., direct the network traffic associated with UE **502** to the unlicensed spectrum).

(73) At step **655**, SM-PCF **514** sends an SM policy update message to SMF **510**/UPF **512**, which indicates the migration of network traffic to the unlicensed spectrum. At step **660**, SMF **510**/UPF **512** sends an N1/N2 message to AMF **508** regarding the same. At step **665**, AMF **508** forwards the N2 message to gNodeB **504**, which then sends an RRC reconfiguration to UE **502** at step **670**.

(74) At step **675**, UE **502**, based on the RRC reconfiguration, connects to cell-UL on F2 radiofrequency. As follows, at step **680**, network traffic associated with UE **502** is now directed to

the unlicensed spectrum.

(75) While example flow **600** includes the migration of network traffic triggered by the volume of network data exceeding a quota threshold (e.g., Trigger 4 as illustrated with respect to FIG. 3), other types of triggers can be applied.

(76) FIG. 7 illustrates an example flow **700** of reverting network traffic from an unlicensed spectrum to a licensed spectrum, according to some aspects of the present disclosure. Network environment for example flow **700** comprises UE **702** (can be the same as UE **302/502**), gNodeB **704** (can be the same as gNodeB **304/504**), AMF **708**, SMF **710/UPF 712**, SM-PCF **714**, and UDM **716**.

(77) At step **720**, gNodeB **704** can be configured to support both licensed and unlicensed frequencies (e.g., cell-L on F1 radio frequency and cell-UL on F2 radiofrequency). At step **725**, UE **702** connects to cell-L on F1 radio frequency via gNodeB **704**.

(78) At step **730**, UE registration is completed between UE **702** and AMF **708**. At step **635**, AMF **708** and UMD **716** confirm the subscription of UE **702** and process authentication of UE **702**. At step **740**, UE **702** sends an internet PDU session over the licensed spectrum to SMF **710/UPF 712**.

(79) At step **745**, based on a preconfigured rule, AMF **708** moves a portion of or all traffic associated with UE **702** to an unlicensed spectrum (e.g., based on Trigger 3 as illustrated in FIG. 3). While example flow **700** includes the migration of the network traffic to the unlicensed spectrum triggered by a preconfigured rule, other types of triggers can be applied.

(80) At step **750**, AMF **708** sends an N2 message to gNodeB **704** indicating the radiofrequency change. At step **755**, gNodeB **704** sends an RRC reconfiguration to UE **702**. As follows, UE **702**, based on the RRC reconfiguration, connects to cell-UL on F2 radiofrequency via gNodeB **704** at step **760**. Thus, network traffic associated with UE **602** is now directed on the unlicensed spectrum to SMF **710/UPF 712** at step **765**.

(81) In some examples, at step **770**, based on a preconfigured rule, all traffic previously moved to the unlicensed spectrum can be moved back to a licensed spectrum. For example, the preconfigured rule can define a schedule for what time the traffic needs to be directed to a licensed spectrum or an unlicensed spectrum. Further, the preconfigured rule can define which and what portion of the network traffic needs to be migrated to the unlicensed spectrum or reverted to the licensed spectrum.

(82) At step **775**, AMF **708** sends an N2 message indicating the radiofrequency change (e.g., from F2 radio frequency to F1 radio frequency) to gNodeB **704**. At step **780**, gNodeB **704** can send an RRC reconfiguration regarding the same to UE **702**. As follows, UE **702** connects to cell-L on F1 radio frequency via gNodeB **704** at step **785**. Thus, network traffic associated with UE **702** is directed on the licensed spectrum at step **790**.

(83) While example flow **700** includes the reverting process, which is triggered by a preconfigured rule, any other types of triggers for reverting network traffic back to the licensed spectrum from the unlicensed spectrum can be used and falls within the scope of the present disclosure. can be applied.

(84) FIG. 8 illustrates a flow chart of a network traffic migration process/method **800** for migrating network traffic from a licensed spectrum to an unlicensed spectrum within the same RAT, according to some aspects of the present disclosure. Although the example method **800** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the method **700**. In other examples, different components of an example device or system that implements the method **800** may perform functions at substantially the same time or in a specific sequence. The process of FIG. 8 may be implemented by a network controller. Such network controller can be any one of fog nodes **162** (e.g., routers, controllers, cameras, access points, gateways), gNodeB (RAN) **220**, an NF inside core network **230**, AMF **306**, SMF **332B**, etc. It should be noted that such network controller

may have one or more memories having computer-readable instructions stored therein and one or more associated processors configured to execute the computer-readable instructions to perform steps of method **800** as described below.

(85) At step **810**, method **800** includes identifying a user device connected to a cellular wireless access technology over a licensed spectrum. For example, an enhanced roaming system can identify UE **202** as illustrated in FIG. 2, which is connected to a cellular wireless access technology (e.g., via RAN **204** as illustrated in FIG. 2) over a licensed spectrum (e.g., 2.5 GHZ).

(86) At step **820**, method **800** includes determining whether a condition for switching network traffic associated with the user device to an unlicensed spectrum is triggered. The condition can be any one of the triggers described above with reference to FIGS. 4A-D, a threshold quota described with reference to FIG. 6, preconfigured triggers such as a schedule for moving traffic from a licensed spectrum to an unlicensed spectrum as described with reference to FIG. 7, etc. In one instance, method **800** includes receiving, from a network element, a usage report that includes a volume of the network traffic of the user device to compare to the capacity threshold of the licensed spectrum. For example, an enhanced roaming system can receive, from a network element of first cell **230A** (e.g., SMF **232A**, UPF **234A**, PCF **236A**, NRF **238A**, etc.), a usage report that includes a volume of the network traffic of UE **202** to compare to the capacity threshold of first cell **230A** in order to determine whether the network traffic of UE **202** should be moved to the unlicensed spectrum or not.

(87) If at step **820**, the network controller determines that the condition is not triggered (is not met), the network controller continues to steer network traffic associated with the user device (e.g., UE **202**) over the licensed spectrum. However, if at step **820**, the network controller determines that the condition for switching network traffic associated with the user is triggered, then at step **830**, method **800** includes determining an unlicensed spectrum to move the network traffic to. The unlicensed spectrum can be within a same cell as the licensed spectrum (as described with reference to FIGS. 5-7) or can be in a different cell compared to a cell in which the licensed spectrum is (as described with reference to FIGS. 3 and 4A-4D).

(88) At step **840**, method **800** includes migrating at least a portion of the network traffic of the user device to the unlicensed spectrum in a similar manner as described above with reference to FIGS. 3-7. to the second cell on the unlicensed spectrum while maintaining network connectivity of the user device over the cellular wireless access technology. In one example, such migrating may occur by modifying a User Equipment Route Selection Policy (URSP) for the user device to migrate the portion of the network traffic to the unlicensed spectrum. In another example, such migrating may include transmitting a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the unlicensed spectrum.

(89) As described above with reference to FIGS. 6 and 7, network traffic that is moved to an unlicensed spectrum may be reverted back (migrated) back to the licensed spectrum. Accordingly, at step **850**, method **8** includes determining whether a condition for reverting network traffic back to the licensed spectrum is triggered (met). If not, the network controller continues to steep the network traffic for the user device over the unlicensed spectrum until the condition for reverting back to the licensed spectrum is met. Once the reverting back condition is triggered, at step **860**, the method **800** includes migrating the portion of the network traffic that was moved to the unlicensed spectrum at step **840**, back to the licensed spectrum. In one example, such migrating may occur by modifying a User Equipment Route Selection Policy (URSP) for the user device to migrate the portion of the network traffic back to the licensed spectrum from the unlicensed spectrum. In another example, such migrating may include transmitting a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the unlicensed spectrum.

(90) With examples of network traffic steering between licensed and unlicensed spectrum within a 3GPP service described above with reference to FIGS. 1-8, as alternative methods of throttling, the

disclosure now turns to example systems and components that can be utilized any of the network components described above with reference to FIGS. 1, 2, 3, and 5 including a network controller that may implement the process of FIG. 8.

(91) FIG. 9 illustrates an example computing system, according to some aspects of the present disclosure. FIG. 9 illustrates an example computing system **900** including components in electrical communication with each other using a connection **905** upon which one or more aspects of the present disclosure can be implemented. Connection **905** can be a physical connection via a bus, or a direct connection into processor **910**, such as in a chipset architecture. Connection **905** can also be a virtual connection, networked connection, or logical connection.

(92) In some embodiments computing system **900** is a distributed system in which the functions described in this disclosure can be distributed within a datacenter, multiple datacenters, a peer network, etc. In some embodiments, one or more of the described system components represents many such components each performing some or all of the function for which the component is described. In some embodiments, the components can be physical or virtual devices.

(93) Example system **900** includes at least one processing unit (CPU or processor) **910** and connection **905** that couples various system components including system memory **915**, such as read only memory (ROM) **920** and random access memory (RAM) **925** to processor **910**.

Computing system **900** can include a cache of high-speed memory **912** connected directly with, in close proximity to, or integrated as part of processor **910**.

(94) Processor **910** can include any general purpose processor and a hardware service or software service, such as services **932**, **934**, and **936** stored in storage device **930**, configured to control processor **910** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **910** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(95) To enable user interaction, computing system **900** includes an input device **945**, which can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech, etc. Computing system **900** can also include output device **935**, which can be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems can enable a user to provide multiple types of input/output to communicate with computing system **900**.

Computing system **900** can include communications interface **940**, which can generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(96) Storage device **930** can be a non-volatile memory device and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs), read only memory (ROM), and/or some combination of these devices.

(97) The storage device **930** can include software services, servers, services, etc., that when the code that defines such software is executed by the processor **910**, it causes the system to perform a function. In some embodiments, a hardware service that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **910**, connection **905**, output device **935**, etc., to carry out the function.

(98) FIG. 10 illustrates an example of a network device, according to some aspects of the present disclosure. FIG. 10 illustrates an example network device **1000** suitable for performing switching, routing, load balancing, and other networking operations. Network device **1000** includes a central processing unit (CPU) **1004**, interfaces **1002**, and a bus **1010** (e.g., a PCI bus). When acting under

the control of appropriate software or firmware, the CPU **1004** is responsible for executing packet management, error detection, and/or routing functions. The CPU **1004** preferably accomplishes all these functions under the control of software including an operating system and any appropriate applications software. CPU **1004** may include one or more processors **1008**, such as a processor from the INTEL X86 family of microprocessors. In some cases, processor **1008** can be specially designed hardware for controlling the operations of network device **1000**. In some cases, a memory **1006** (e.g., non-volatile RAM, ROM, etc.) also forms part of CPU **1004**. However, there are many different ways in which memory could be coupled to the system.

(99) The interfaces **1002** are typically provided as modular interface cards (sometimes referred to as “line cards”). Generally, they control the sending and receiving of data packets over the network and sometimes support other peripherals used with the network device **1000**. Among the interfaces that may be provided are Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, and the like. In addition, various very high-speed interfaces may be provided such as fast token ring interfaces, wireless interfaces, Ethernet interfaces, Gigabit Ethernet interfaces, ATM interfaces, HSSI interfaces, POS interfaces, FDDI interfaces, WIFI interfaces, 3G/4G/5G cellular interfaces, CAN BUS, LoRA, and the like. Generally, these interfaces may include ports appropriate for communication with the appropriate media. In some cases, they may also include an independent processor and, in some instances, volatile RAM. The independent processors may control such communications intensive tasks as packet switching, media control, signal processing, crypto processing, and management. By providing separate processors for the communications intensive tasks, these interfaces allow the master CPU **1004** to efficiently perform routing computations, network diagnostics, security functions, etc.

(100) Although the system shown in FIG. **10** is one specific network device of the present technology, it is by no means the only network device architecture on which the present technology can be implemented. For example, an architecture having a single processor that handles communications as well as routing computations, etc., is often used. Further, other types of interfaces and media could also be used with the network device **1000**.

(101) Regardless of the network device's configuration, it may employ one or more memories or memory modules (including memory **1006**) configured to store program instructions for the general-purpose network operations and mechanisms for roaming, route optimization and routing functions described herein. The program instructions may control the operation of an operating system and/or one or more applications, for example. The memory or memories may also be configured to store tables such as mobility binding, registration, and association tables, etc. Memory **1006** could also hold various software containers and virtualized execution environments and data.

(102) The network device **1000** can also include an application-specific integrated circuit (ASIC), which can be configured to perform routing and/or switching operations. The ASIC can communicate with other components in the network device **1000** via the bus **1010**, to exchange data and signals and coordinate various types of operations by the network device **1000**, such as routing, switching, and/or data storage operations, for example.

(103) For clarity of explanation, in some instances the various embodiments may be presented as including individual functional blocks including functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software.

(104) In some embodiments the computer-readable storage devices, media, and memories can include a cable or wireless signal containing a bit stream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

(105) Methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer readable media. Such

instructions can comprise, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, or source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.

(106) Devices implementing methods according to these disclosures can comprise hardware, firmware and/or software, and can take any of a variety of form factors. Some examples of such form factors include general purpose computing devices such as servers, rack mount devices, desktop computers, laptop computers, and so on, or general purpose mobile computing devices, such as tablet computers, smart phones, personal digital assistants, wearable devices, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

(107) The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are means for providing the functions described in these disclosures.

(108) Although a variety of examples and other information was used to explain aspects within the scope of the appended claims, no limitation of the claims should be implied based on particular features or arrangements in such examples, as one of ordinary skill would be able to use these examples to derive a wide variety of implementations. Further and although some subject matter may have been described in language specific to examples of structural features and/or method steps, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to these described features or acts. For example, such functionality can be distributed differently or performed in components other than those identified herein. Rather, the described features and steps are disclosed as examples of components of systems and methods within the scope of the appended claims.

(109) Claim language reciting “at least one of” refers to at least one of a set and indicates that one member of the set or multiple members of the set satisfy the claim. For example, claim language reciting “at least one of A and B” means A, B, or A and B.

Claims

1. A method comprising: identifying a user device connected to a cellular wireless access technology over an existing spectrum, the existing spectrum being one of a licensed spectrum or an unlicensed spectrum, the unlicensed spectrum being within a same cell or different cell as the licensed spectrum; receiving information from multiple control-plane network elements indicative of network utilization parameters for the user device over the existing spectrum; determining whether a condition for switching network traffic associated with the user device to an alternative spectrum is triggered based at least in part on the information received, the alternative spectrum being different than the existing spectrum; in response to determining that the condition is triggered, determining the alternative spectrum to move the network traffic to, wherein the alternative spectrum is the licensed spectrum if the existing spectrum is the unlicensed spectrum or the alternative spectrum is the unlicensed spectrum if the existing spectrum is the licensed spectrum; and migrating at least a portion of the network traffic to the alternative spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.
2. The method of claim 1, wherein the condition includes a capacity threshold of the licensed

spectrum, and the network traffic is migrated to the alternative spectrum if the capacity threshold of the existing spectrum is reached.

3. The method of claim 2, wherein the existing spectrum is the licensed spectrum and the alternative spectrum is the unlicensed spectrum.

4. The method of claim 2, wherein the utilization parameters include a volume of the network traffic of the user device to compare to the capacity threshold.

5. The method of claim 1, further comprising: modifying a User Equipment Route Selection Policy (URSP) to migrate the portion of the network traffic to the alternative spectrum.

6. The method of claim 1, further comprising: determining, after migrating the portion of the network traffic to the alternative spectrum, whether a second condition for switching the network traffic back to the existing spectrum is met; and migrating the portion of the network traffic back to the existing spectrum from the alternative spectrum.

7. The method of claim 5, further comprising: transmitting a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the alternative spectrum.

8. A network controller comprising: one or more memories having computer-readable instructions stored therein; and one or more processors configured to execute the computer-readable instructions to: identify a user device connected to a cellular wireless access technology over an existing spectrum, the existing spectrum being one of a licensed spectrum or an unlicensed spectrum, the unlicensed spectrum being within a same cell or different cell as the licensed spectrum; receive information from multiple control-plane network elements indicative of network utilization parameters for the user device over the existing spectrum; determine whether a condition for switching network traffic associated with the user device to an alternative spectrum is triggered based at least in part on the information received, the alternative spectrum being different than the existing spectrum; in response to determining that the condition is triggered, determine the alternative spectrum to move the network traffic to, wherein the alternative spectrum is the licensed spectrum if the existing spectrum is the unlicensed spectrum or the alternative spectrum is the unlicensed spectrum if the existing spectrum is the licensed spectrum; and migrate at least a portion of the network traffic to the alternative spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

9. The network controller of claim 8, wherein the condition includes a capacity threshold of the licensed spectrum, and the network traffic is migrated to the alternative spectrum if the capacity threshold of the existing spectrum is reached.

10. The network controller of claim 9, wherein the existing spectrum is the licensed spectrum and the alternative spectrum is the unlicensed spectrum.

11. The network controller of claim 9, wherein the utilization parameters include a volume of the network traffic of the user device to compare to the capacity threshold.

12. The network controller of claim 8, wherein the one or more processors are configured to execute the computer-readable instructions to modify a User Equipment Route Selection Policy (URSP) to migrate the portion of the network traffic to the alternative spectrum.

13. The network controller of claim 8, wherein the one or more processors are configured to execute the computer-readable instructions to: determine, after migrating the portion of the network traffic to the alternative spectrum, whether a second condition for switching the network traffic back to the existing spectrum is met; and; and the portion of the network traffic back to the existing spectrum from the alternative spectrum.

14. The network controller of claim 8, wherein the one or more processors are configured to execute the computer-readable instructions to transmit a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the alternative spectrum.

15. One or more non-transitory computer-readable media comprising computer-readable

instructions, which when executed by one or more processors of a network controller, cause the network controller to: identify a user device connected to a cellular wireless access technology over an existing spectrum, the existing spectrum being one of a licensed spectrum or an unlicensed spectrum, the unlicensed spectrum being within a same cell or different cell as the licensed spectrum; receive information from multiple control-plane network elements indicative of network utilization parameters for the user device over the existing spectrum; determine whether a condition for switching network traffic associated with the user device to an alternative spectrum is triggered based at least in part on the information received, the alternative spectrum being different than the existing spectrum; in response to determining that the condition is triggered, determine the alternative spectrum to move the network traffic to, wherein the alternative spectrum is the licensed spectrum if the existing spectrum is the unlicensed spectrum or the alternative spectrum is the unlicensed spectrum if the existing spectrum is the licensed spectrum; and migrate at least a portion of the network traffic to the alternative spectrum while maintaining network connectivity of the user device over the cellular wireless access technology.

16. The one or more non-transitory computer-readable media of claim 15, wherein the condition includes a capacity threshold of the licensed spectrum, and the network traffic is migrated to the alternative spectrum if the capacity threshold of the existing spectrum is reached.

17. The one or more non-transitory computer-readable media of claim 16, wherein the utilization parameters include a volume of the network traffic of the user device to compare to the capacity threshold.

18. The one or more non-transitory computer-readable media of claim 15, wherein the execution of the computer-readable instructions by the one or more processors, further cause the network controller to modify a User Equipment Route Selection Policy (URSP) to migrate the portion of the network traffic to the alternative spectrum.

19. The one or more non-transitory computer-readable media of claim 15, wherein the execution of the computer-readable instructions by the one or more processors, further cause the network controller to: determine, after migrating the portion of the network traffic to the alternative spectrum, whether a second condition for switching the network traffic back to the existing spectrum is met; and; and the portion of the network traffic back to the existing spectrum from the alternative spectrum.

20. The one or more non-transitory computer-readable media of claim 15, wherein the execution of the computer-readable instructions by the one or more processors, further cause the network controller to transmit a Radio Resource Control (RRC) connection reconfiguration message to the user device to migrate the portion of the network traffic to the alternative spectrum.
